Ultra Low Power +3.3V Dual MIL-STD-1553 Transceiver

Model: BU-67401L

Data Sheet

World's lowest power dual MIL-STD-1553 transceiver in a 7 mm x 7 mm package*, operating from a +3.3V power supply voltage. This transceiver operates with "Harris" type encoders/decoders and features an extremely small footprint in a 48-pad Leadless Plastic Chip Carrier (LPCC).*

Features

- World's Lowest Power Dual MIL-STD-1553 Transceiver
- Options for MIL-STD-1553A/B, MIL-STD-1760, & McAir Compatibility
- -55°C to +125°C Operation
- Very Small 7 mm x 7 mm (0.28 in x 0.28 in) Body & Footpint Size
- Requires Only +3.3V Power Supply
- Small 48-Pad Leadless Plastic Chip Carrier (LPCC) Package

For more information: www.ddc-web.com/BU-67401L

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MIL-STD-1553 | ARINC 429 | Fibre Channel

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ULTRA LOW POWER +3.3 VOLT MIL-STD-1553 DUAL TRANSCEIVER

BU-67401L DATA SHEET

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Revision	Date	Pages	Description
A	Feb., 2012	All	Initial Release
B	March, 2012	14	Updated figure 5.
C	May, 2012	6	Update Power Distribution information in Table 1
D	June, 2012	16	Added “R” Solder Pad Type option to Ordering Information

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1 PREFACE

This data sheet uses typographical conventions to assist the reader in understanding the content. This section will define the text formatting used in the rest of the data sheet.

1.1 Text Usage

- **BOLD**–indicates important information and table, figure, and chapter references.
- ***BOLD ITALIC***–designates DDC Part Numbers.
- `Courier New`–indicates code examples.
- <…> - indicates user-entered text or commands.

1.2 Special Handling and Cautions

Warnings: Turn off power to the computer hardware and unplug from wall.

Ensure that standard ESD precautions are followed. As a minimum, one hand should be grounded to the power supply in order to equalize the static potential.

Do not store disks in environments exposed to excessive heat, magnetic fields or radiation.

1.3 Trademarks

All trademarks are the property of their respective owners.

1.4 What is included in this data sheet?

This data sheet contains a complete description of component.

1.5 Technical Support

In the event that problems arise beyond the scope of this data sheet, you can contact DDC by the following:

US Toll Free Technical Support:
1-800-DDC-5757, ext. 7771

Outside of the US Technical Support:
1-631-567-5600, ext. 7771

Fax:
1-631-567-5758 to the attention of DATA BUS Applications

DDC Website:
www.ddc-web.com/ContactUs/TechSupport.aspx

Please note that the latest revisions of Software and Documentation are available for download at DDC's Web Site, www.ddc-web.com.

2 OVERVIEW

The ***BU-67401L*** is a MIL-STD-1553 dual transceiver that operates from a +3.3V power supply voltage. It operates with Harris type encoder/decoders and features an ultra low profile and small footprint in a 48-pad Leadless Plastic Chip Carrier package (LPCC). The LPCC package provides an integral, exposed heatsink on the package bottom, which allows 100% duty cycle operation at +125°C when the heatsink is soldered to an appropriately constructed PCB.

The receiver section of the ***BU-67401L*** accepts Manchester II data from a MIL-STD-1553 Data Bus and produces TTL level signals at its outputs: RX_DATA_OUT and $\overline{\text{RX\_DATA\_OUT}}$. These outputs represent positive and negative excursions of the input data signals beyond an internally fixed threshold level. An external STROBE input enables or disables the receiver's outputs.

The transmitter section accepts bipolar TTL signal data at its TX_DATA_IN and $\overline{\text{TX\_DATA\_IN}}$ inputs and produces Manchester II data at the TXOUT and $\overline{\text{TXOUT}}$ outputs.

An external input, TX_INHIBIT, takes priority over the transmitter inputs and disables the respective transmitter when asserted to logic "1". The ultra small size, +3.3V power supply voltage, and options for compliance with MIL-STD-1553A, 1553B, 1760, and McAir simplify engineering design, making it an excellent choice for interfacing with any MIL-STD-1553 system.

2.1 Features

- The World’s Lowest Power Dual MIL-STD-1553 Transceiver
- Options for MIL-STD-1553A/B, MIL-STD-1760, and McAir Compatible
- -55°C to +125°C Operation
- 0.28 x 0.28 inches (7 x 7 mm) body and Footprint Size
- Requires Only +3.3V Power Supply
- Small 48-Pad LPCC Package
- Dual Transceiver

Applications Include:

- Mission Computers
- Digital Data Recorders
- LRUs

- Radios/Modems
- Displays
- Ground Vehicles
- Commercial Aerospace
- Radar Systems/Situational Awareness

Figure 1. BU-67401L Transceiver

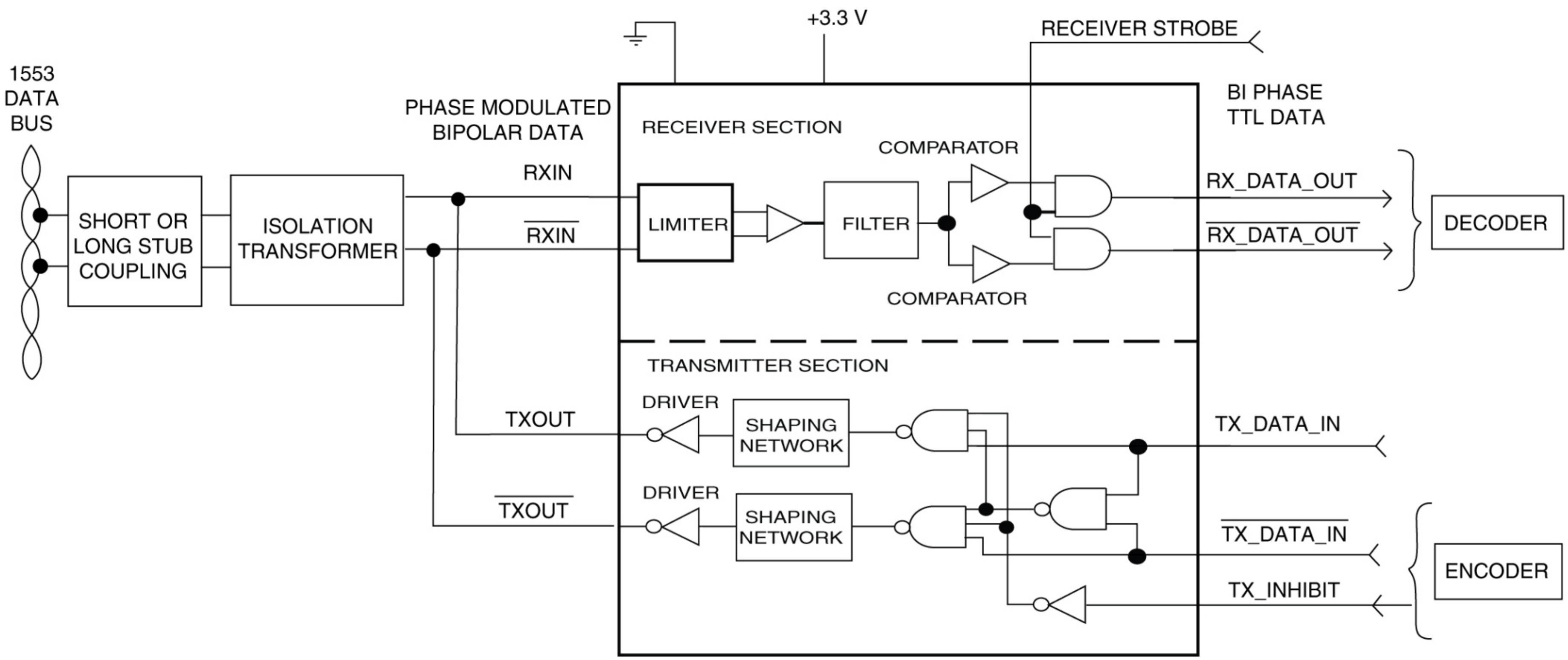


Figure 2. BU-67401L Block Diagram

(One Channel Shown)

PARAMETER	MIN	TYP	MAX	UNITS
ABSOLUTE MAXIMUM RATING				
Supply Voltage				
+3.3 V	-0.3	3.3	4.5	Vdc
Receiver				
Input Voltage (at device pins) (Note 13)			±3.5	V
Logic				
Voltage Input Range	-0.3		Vcc + 0.3	V
POWER SUPPLY REQUIREMENTS				
Voltages/Tolerances				
+3.3V	3.14	3.3	3.46	V
Current Drain (Note 10)				
• Idle			26	mA
• 25% Transmitter Duty Cycle (1 Ch. Transmitting)			200	mA
• 50% Transmitter Duty Cycle (1 Ch. Transmitting)			330	mA
• 100% Transmitter Duty Cycle (1 Ch. Transmitting)			610	mA
POWER DISSIPATION				
Vcc = +3.3V (Note 10)				
• Idle			0.08	W
• 25% Transmitter Duty Cycle (1 Ch. Transmitting)			0.19	W
• 75% Transmitter Duty Cycle (1 Ch. Transmitting)			0.29	W
• 100% Transmitter Duty Cycle (1 Ch. Transmitting)			0.5	W
RECEIVER				
Differential Input Resistance (Notes 1 – 6, Note 12)	5			kΩ
Differential Input Capacitance (Notes 1 – 6, Note 12)			10	pF
Threshold Voltage, Transformer Coupled, Measured on Stub (Note 7)	.200		.860	Vp-p
Common-Mode Voltage (Note 8)			10	Vpeak
TRANSMITTER				
Differential Output Voltage				
• Direct Coupled Across 35Ω, Measured on Bus	6	7.4	9	Vp-p
• Transformer Coupled Across 70Ω, Measured on Stub (Note 9)				
1553/1760	20	22	27	Vp-p
McAir	18		27	Vp-p
Output Noise, Differential (Direct Coupled)			5	mVrms, diff
Output Offset Voltage, Transformer Coupled Across 70Ω	-250		+250	mVp, diff
Rise/Fall Time				
1553/1760	100	150	300	ns
McAir	200	220	300	ns

Table 1. BU-67401L Specification Table

Table 1. BU-67401L Specification Table

PARAMETER	MIN	TYP	MAX	UNITS
LOGIC				
V_{IH}	0.7 x Vcc			V
V_{IL}			0.3 x Vcc	V
I_{IH} (TX_DATA_IN, $\overline{\text{TX\_DATA\_IN}}$, TX_INHIBIT)	20		100	µA
I_{IH} (STROBE)			0	µA
I_{IL} (TX_DATA_IN, $\overline{\text{TX\_DATA\_IN}}$, TX_INHIBIT)	0			µA
I_{IL} (STROBE)	-100		-20	µA
V_{OH} (I_{OH} = max)	Vcc – 0.4			V
V_{OL} (I_{OL} = min)			0.4	V
I_{OL}	2.0			mA
I_{OH}			-2.0	mA
THERMAL				
Thermal Resistance, Junction-to-Heatsink (θ_{JS})		5.6	9	°C/W
Thermal Resistance, Junction-to-Board(θ_{JB}) (Note 11) Heat Sink Soldered to PC Board (2s2p – JESD51-5)		13	17	°C/W
Operating Case Bottom Temperature	-55		+125	°C
Operating Junction Temperature	-55		+150	°C
Storage Temperature	-65		+150	°C
Pad Temperature (soldering, 10 sec.)			+300	°C
PHYSICAL CHARACTERISTICS				
Moisture Sensitivity Level	MSL-2			—
Package Body Size 48-pad LPCC	0.280 x 0.280 x 0.040 (7.1x 7.1 x 1.0)			in. (mm)
Weight	.0054 (.15)			oz. (g)

Table 1. BU-67401L Specification Table

PARAMETER	MIN	TYP	MAX	UNITS

Table 1 Notes:

Notes 1 through 6 are applicable to the Receiver Differential Resistance and Differential Capacitance specifications:

1. Specifications include both transmitter and receiver (assumed tied together externally without a transformer).
2. Impedance parameters are specified directly between pads TXOUT/RXIN and $\overline{\text{TXOUT}}$ / $\overline{\text{RXIN}}$ of the package.
3. It is assumed that all power and ground inputs to the package are connected.
4. The specifications are applicable for both un-powered and powered conditions.
5. The specifications assume a 1.5-volt rms balanced differential, sinusoidal input. The applicable frequency range is 75 kHz to 1 MHz.
6. Minimum resistance and maximum capacitance parameters are guaranteed over the operating range, but are not tested.
7. The Threshold Level, as referred to in this specification, is meant to be the maximum peak-to-peak voltage (measured on the stub) that can be applied to the receiver's input without causing the output to change from the OFF state.
8. Assumes a common mode voltage within the frequency range of dc to 2 MHz, applied to pins of the isolation transformer on the stub side (either direct or transformer coupled), and referenced to hybrid ground. Transformer must be a DDC recommended transformer.
9. MIL-STD-1760 requires minimum output voltage of 20Vp-p on the stub connection.
10. Current drain and power dissipation specifications are preliminary and subject to change. Power dissipation of the transceiver is defined as the total power into the device minus the power dissipated in the load.
11. Thermal resistance specs are preliminary and subject to change. Junction-to-board thermal resistance is measured in accordance with JEDEC standard JESD51-8. The 16 thermal vias connecting the LPCC heatsink to PCB internal plane are in accordance with JEDEC JESD51-5 (2s2p). Each via is 0.3 mm diameter with 0.035 mm plating. Please refer to JEDEC standard JESD51-5 for detailed PCB construction.
12. To calculate the loading effect on the stub side of the "long stub" isolation transformer, multiply the input resistance by 4.28 (Isolation Transformer ratio of 2.07^2) and divide the input capacitance by 4.28. For "short stub" isolation transformers, multiply the input resistance by 7.02 (Isolation Transformer ratio of 2.65^2) and divide the input capacitance by 7.02.
13. Assuming the use of isolation transformers with the turns ratios shown in Figure 4 and in the absence of common mode signal on the 1553 stub, this equates to a stub voltage of 29.0 $\text{Volts}_{\text{PK-to-PK}}$ transformer-coupled, or 37.1 $\text{Volts}_{\text{PK-to-PK}}$ direct-coupled.

3 INTRODUCTION

The ***BU-67401L*** is a dual transmitter and receiver packaged in a 48-pad Leadless Plastic Chip Carrier (LPCC). It is directly compatible with Harris type encoder/decoders and has internal (factory preset) threshold levels. The transceiver only requires +3.3V power and conforms to MIL-STD-1553A/B, MIL-STD-1760, and McAir.

Figure 4 illustrates the connection between a ***BU-67401L*** transceiver and a MIL-STD-1553 data bus. Used with an isolation transformer with the appropriate turns ratio, it can be either direct coupled (short stub) or transformer coupled (long stub) to the 1553 data bus.

3.1 TRANSMIT OPERATING MODE

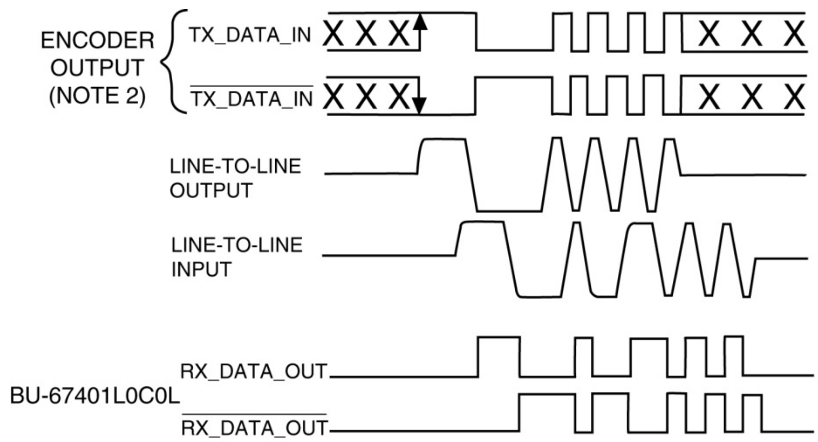
The transmitter section accepts encoded TTL data and converts it to bi-phase Manchester II form using a waveshaping network and driver circuits. The driver outputs TXOUT and $\overline{\text{TXOUT}}$ are transformer isolated from the Data Bus.

The transmitter output terminals can be put into a high impedance state by setting TX_INHIBIT high, or asserting TX_DATA_IN and $\overline{\text{TX\_DATA\_IN}}$ to the same logic level.

3.2 RECEIVER OPERATING MODE

The receiver section accepts data from a MIL-STD-1553 Data Bus when coupled to the Data Bus as shown in Figure 4. This data is converted to TTL levels and provided as RX_DATA_OUT and $\overline{\text{RX\_DATA\_OUT}}$.

When STROBE is high, data passes through the receiver to RX_DATA_OUT and $\overline{\text{RX\_DATA\_OUT}}$. Applying a low to STROBE disables the receiver output terminals. As illustrated in Figure 3, the receiver in the ***BU-67401L*** provides compatibility with Harris type decoders.



Notes:
(1) TX_DATA_IN and RX_DATA_OUT are TTL signals.
(2) TX_DATA_IN inputs must be at the same logic level when not transmitting.
(3) LINE-TO-LINE output voltage is measured between TX_DATA_OUT and $\overline{\text{TX\_DATA\_OUT}}$.
(4) LINE-TO-LINE input voltage is measured on the Data Bus.

Figure 3. Waveforms for Harris Type Encoder/Decoder

3.3 WAVEFORMS

Figure 3 illustrates the ***BU-67401L*** with Harris type decoder interface. Note that TX_DATA_IN and $\overline{\text{TX\_DATA\_IN}}$ inputs must be complementary waveforms with a 50% duty cycle.

3.4 1553 BUS INTERFACE AND LAYOUT CONSIDERATIONS

Figure 4 illustrates the interface between the ***BU-67401L*** and a MIL-STD-1553 bus. Connections for both direct (short stub) and transformer (long stub) coupling, as well as the peak-to-peak voltage levels at various points (when transmitting), are indicated in the diagram.

Note that the BU-67401L has two pads for each of the TXOUT phases. Both pads of the same signal phase must be connected together on the PCB to assure adequate current carrying capacity.

The center tap of the primary winding (the side of the transformer that connects to the ***BU-67401L***) must be directly connected to ground. Additionally, during transmission, large currents flow from the transceiver power supply through the TX/RX pins into the transformer primaries and then out the center tap into the ground plane. The traces in this path should be sized accordingly and the connections to the ground plane should be as short as possible.

It is recommended that a 10 µF low inductance capacitor (tantalum or ceramic) be placed as close as possible to one of the transceiver power inputs of the ***BU-67401L*** for each transceiver channel.

It is also recommended that a 0.01µF ceramic capacitor be mounted as close as possible and with the shortest leads to each transceiver power input of the ***BU-67401L***.

To achieve its full military temperature range rating, the ***BU-67401L*** needs to be properly heatsinked. The thermal resistance junction-to-board specification (Table 1) allows for 16 thermal vias connecting the heatsink on the package bottom to a 2s2p (2 sided, 2 plane PCB), 3.0" x 4.5". The thermal via construction follows JEDEC standard JESD51-5 with thermal vias connecting the top side mini-plane located under the heatsink to the upper internal plane. The board temperature is determined 1mm from the package in accordance with JEDEC standard JESD51-8.

The 2s2p PCB construction has 2 inner planes of 1 oz copper, 2 outer trace layers of 2 oz copper and has an overall thickness of 1.6 mm, which is divided evenly amongst the 3 dielectrics. The top layer has 12 traces (0.26 mm width each) fanning out from each side of the package. Refer to the appropriate JEDEC standards, and DDC's Application Note AN/B-39 "MIL-STD-1553 LPCC PACKAGED TRANSCEIVERS, A USER'S GUIDE" for further information.

3.5 ISOLATION TRANSFORMERS

In selecting isolation transformers to be used with the ***BU-67401L***, there is a limitation on the maximum amount of leakage inductance. If this limit is exceeded, the transmitter rise and fall times may increase, possibly causing the bus amplitude to fall below the minimum level required by MIL-STD-1553. In addition, an excessive leakage imbalance may result in a transformer dynamic offset that exceeds 1553 specifications.

The maximum allowable leakage inductance is a function of the coupling method. For Transformer Coupled applications, it is a maximum of 5.0 µH. For Direct Coupled applications, it is a maximum of 10.0 µH, and is measured as follows:

The side of the transformer that connects to the ***BU-67401L*** is defined as the "primary" winding. If one side of the primary is shorted to the primary center-tap, the inductance should be measured across the "secondary" (stub side) winding. This inductance must be less than 5.0 µH (Transformer Coupled) and 10.0 µH (Direct Coupled). Similarly, if the other side of the primary is shorted to the primary center-tap, the inductance measured across the "secondary" (stub side) winding must also be less than 5.0 µH (Transformer Coupled) and 10.0 µH (Direct Coupled).

The difference between these two measurements is the "differential" leakage inductance. This value must be less than 1.0 µH (Transformer Coupled) and 2.0 µH (Direct Coupled).

Beta Transformer Technology Corporation (BTTC), a subsidiary of DDC, manufactures transformers in a variety of mechanical configurations with the required turns ratios of 1:2.65 direct coupled, and 1:2.07 transformer coupled. Table 2 provides a listing of these transformers. For further information, contact BTTC at 631-244-7393 or at www.bttc-beta.com.

Table 2. BTTC Transformer for Use with BU-67401L

BTTC Part #	# of Channels, Configuration	Coupling Description	Coupling Ratio (1:X)	Mounting	Max Height	Width (Including Leads)	Length (Including Leads)
MLP-2030	Single	Direct	(1:2.65)	SMT	0.185”	0.4”	0.52”
MLP-2230	Single	Transformer	(1:2.07)	SMT	0.185”	0.4”	0.52”
DSS-3330	Dual (Side-by-Side)	Direct & Transformer	(1:2.65) & (1:2.07)	SMT	0.185”	0.52”	0.675”
DSS-3330-1	Dual (Side-by-Side) (See Note)	Direct & Transformer, with Secondary Center Taps	(1:2.65) & (1:2.07)	SMT	0.185”	0.52”	0.675”
TSM-2230	Dual (Stacked)	Transformer	(1:2.07)	SMT	0.32”	0.4”	0.52”
DLVB-4230	Dual (Stacked)	Transformer	(1:2.07)	SMT	0.165”	0.72”	0.96”
DSS-1630	Dual (Side-by-Side)	Direct & Transformer	(1:2.65) & (1:2.07)	SMT	0.165”	0.72”	0.96”
LVB-4230	Single	Transformer	(1:2.07)	SMT	0.165”	1.125”	0.625”

Note: Since the DSS-3330-1 includes turns ratios for both direct and transformer coupling and brings out the secondary center taps, it provides compliance with the McAir A3818, A5232, and A5690 standards.

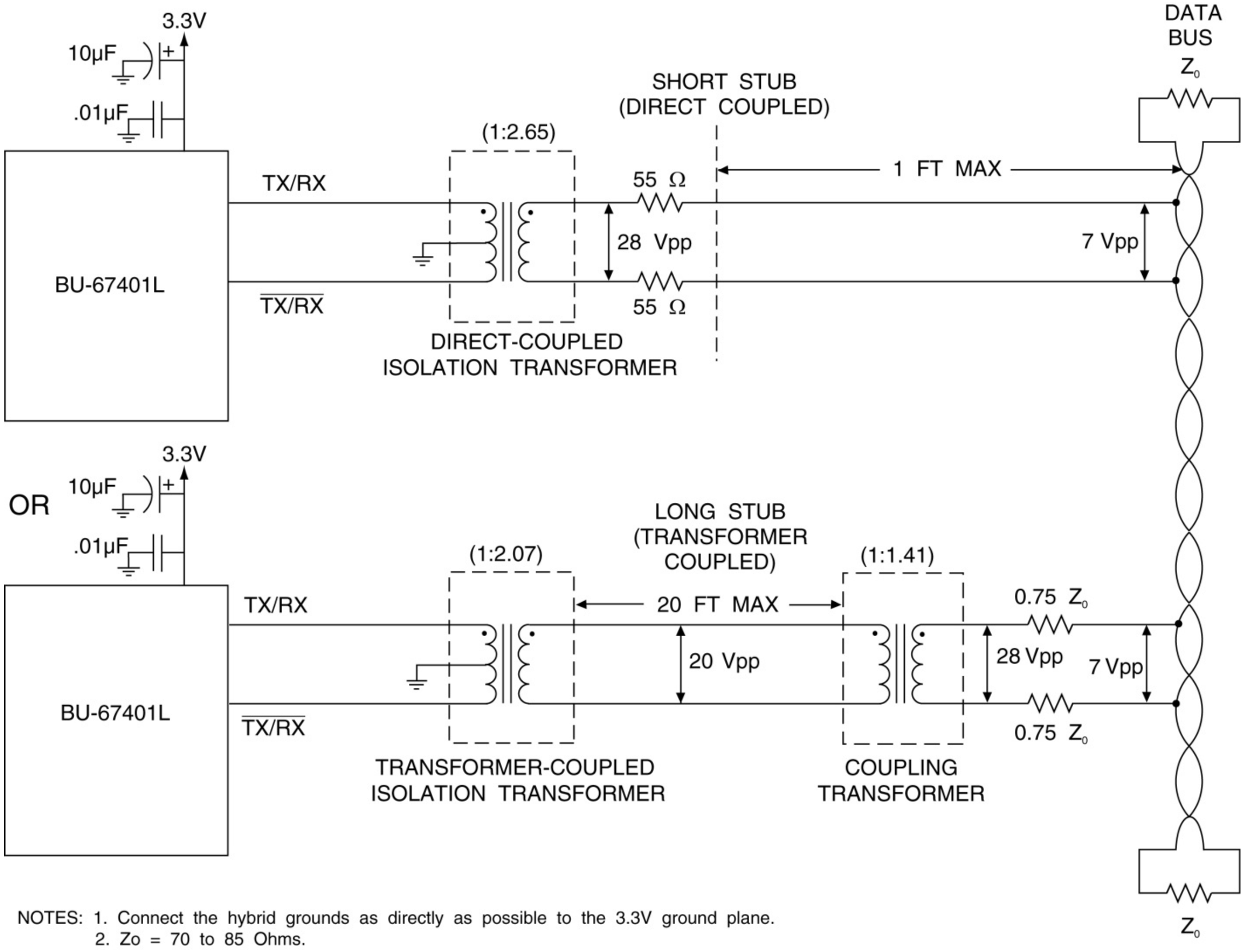


Figure 4. BU-67401L Interface to a MIL-STD-1553 Bus

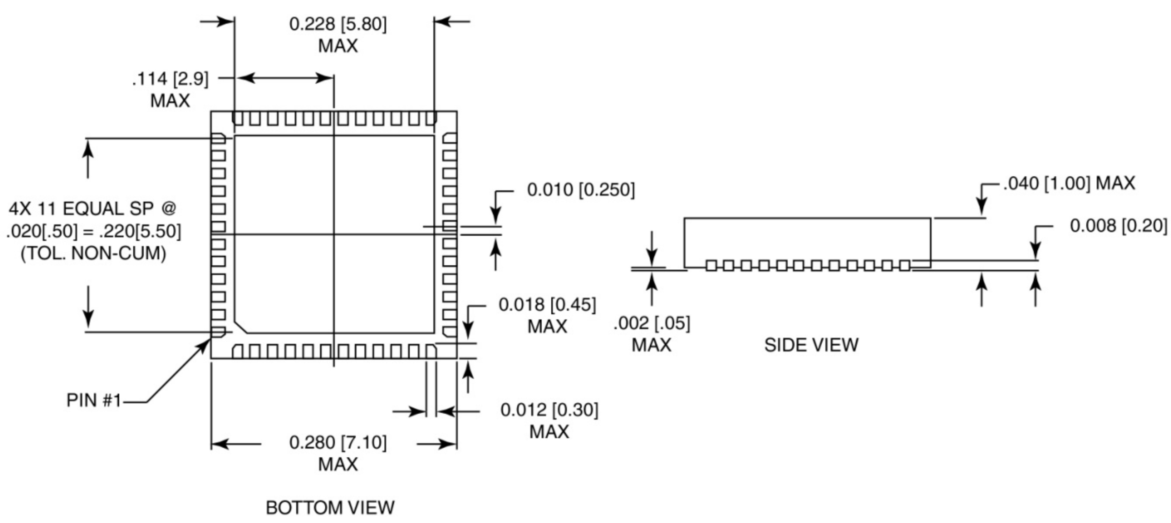


Figure 5. BU-67401L Mechanical Outline

Table 3. BU-67401L Pad Signal Descriptions		
Pad Number	**Name**	**Description**
1	GROUND	GROUND
2	GROUND	GROUND
3	VCC_B	+3.3V Power for Channel B
4	VCC_B	+3.3V Power for Channel B
5	TX_DATA_IN_B	Transmit Data Input (Channel B)
6	$\overline{\text{TX\_DATA\_IN\_B}}$	Inverted Transmit Data Input (Channel B)
7	TX_INHIBIT_B	TX Inhibit Input (Channel B)
8	STROBE_B	Receiver Strobe Input (Channel B)
9	RX_DATA_OUT_B	Receiver Data Output (Channel B)
10	$\overline{\text{RX\_DATA\_OUT\_B}}$	Inverted Receiver Data Output (Channel B)
11	GROUND	GROUND
12	VCC_B	+3.3V Power for Channel B
13	$\overline{\text{RXIN\_B}}$	Inverted Receiver Bus Input (Channel B)
14	RXIN_B	Receiver Bus Input (Channel B)
15	VCC_B	+3.3V Power for Channel B
16	Factory Test Point	DO NOT CONNECT
17	Factory Test Point	DO NOT CONNECT
18	Factory Test Point	DO NOT CONNECT
19	Factory Test Point	DO NOT CONNECT
20	Factory Test Point	DO NOT CONNECT
21	Factory Test Point	DO NOT CONNECT
22	VCC_A	+3.3V Power for Channel A
23	$\overline{\text{RXIN\_A}}$	Inverted Receiver Bus Input (Channel A)
24	RXIN_A	Receiver Bus Input (Channel A)
25	VCC_A	+3.3V Power for Channel A
26	GROUND	GROUND
27	$\overline{\text{RX\_DATA\_OUT\_A}}$	Inverted Receiver Data Output (Channel A)
28	RX_DATA_OUT_A	Receiver Data Output (Channel A)
29	STROBE_A	Receiver Strobe Input (Channel A)
30	TX_INHIBIT_A	TX Inhibit Input (Channel A)
31	TX_DATA_IN_A	Transmit Data Input (Channel A)
32	$\overline{\text{TX\_DATA\_IN\_A}}$	Inverted Transmit Data Input (Channel A)
33	VCC_A	+3.3V Power for Channel A

Table 3. BU-67401L Pad Signal Descriptions		
Pad Number	**Name**	**Description**
34	VCC_A	+3.3V Power for Channel A
35	GROUND	GROUND
36	GROUND	GROUND
37	See Note 1	Configuration Option Input (A)
38	TXOUT_A	Transmitter Bus Output (Channel A)
39	TXOUT_A	Transmitter Bus Output (Channel A)
40	VCC_A	+3.3V Power for Channel A
41	$\overline{\text{TXOUT\_A}}$	Inverted Transmitter Bus Output (Channel A)
42	$\overline{\text{TXOUT\_A}}$	Inverted Transmitter Bus Output (Channel A)
43	TXOUT_B	Transmitter Bus Output (Channel B)
44	TXOUT_B	Transmitter Bus Output (Channel B)
45	VCC_B	+3.3V Power for Channel B
46	$\overline{\text{TXOUT\_B}}$	Inverted Transmitter Bus Output (Channel B)
47	$\overline{\text{TXOUT\_B}}$	Inverted Transmitter Bus Output (Channel B)
48	See Note 1	Configuration Option Input (B)
Heatsink	GROUND	GROUND

Notes:

1. For BU-67401L0C0X connect to Vcc, For BU-67401L0D0X connect to GROUND
2. The heatsink on the package bottom is electrically grounded
3. Connect both pads of each of the TXOUT phases together to ensure sufficient current carrying capability.

4 ORDERING INFORMATION

BU-67401L0 C 0 L - 1 0 2

Bus Amplitude:
- 2 = 20 Vpp minimum (1553/1760 Compliant)
- 0 = 18 Vpp minimum (McAir)

Process Requirements:
- 0 = Standard DDC practices, no Burn-In

Environmental Temperature Options:
- 1 = -55°C to +125°C

Solder Pad Type:
- L = Lead Solder Pad
- R = RoHS Compliant

Rise/Fall Times Option:
- C = 3.3 Volt Dual Transceiver
- D = 3.3 Volt Dual Transceiver, with McAir rise/fall times (Not available with 20 Vpp min amplitude)

Package Identifier:
- L = 48-pad LPCC (Leadless Plastic Chip Carrier)

Notes:
1. Unless otherwise specified, these products contain tin lead solder

Data Device Corporation

Leadership Built on Over 45 Years of Innovation

Military | Commercial Aerospace | Space | Industrial

Data Device Corporation (DDC) is the world leader in the design and manufacture of high-reliability data bus products, motion control, and solid-state power controllers for aerospace, defense, and industrial automation applications. For more than 45 years, DDC has continuously advanced the state of high-reliability data communications and control technology for MIL-STD-1553, ARINC 429, Synchro/Resolver interface, and Solid-State Power Controllers with innovations that have minimized component size and weight while increasing performance. DDC offers a broad product line consisting of advanced data bus technology for Fibre Channel networks; MIL-STD-1553 and ARINC 429 Data Networking cards, components, and software; Synchro/Resolver interface components; and Solid-State Power Controllers and Motor Drives.

Product Families

Data Bus | Synchro/Resolver | Power Controllers | Motor Drives

DDC is a leader in the development, design, and manufacture of highly reliable and innovative military data bus solutions. DDC's Data Networking Solutions include MIL-STD-1553, ARINC 429, and Fibre Channel. Each Interface is supported by a complete line of quality MIL-STD-1553 and ARINC 429 commercial, military, and COTS grade cards and components, as well as software that maintain compatibility between product generations. The Data Bus product line has been field proven for the military, commercial and aerospace markets.

DDC is also a global leader in Synchro/Resolver Solutions. We offer a broad line of Synchro/Resolver instrument-grade cards, including angle position indicators and simulators. Our Synchro/Resolver-to-Digital and Digital-to-Synchro/Resolver microelectronic components are the smallest, most accurate converters, and also serve as the building block for our card-level products. All of our Synchro/Resolver line is supported by software, designed to meet today's COTS/MOTS needs. The Synchro/Resolver line has been field proven for military and industrial applications, including radar, IR, and navigation systems, fire control, flight instrumentation/simulators, motor/motion feedback controls and drivers, and robotic systems.

As the world’s largest supplier of Solid-State Power Controllers (SSPCs) and Remote Power Controllers (RPCs), DDC was the first to offer commercial and fully-qualified MIL-PRF-38534 and Class K Space-level screening for these products. DDC’s complete line of SSPC and RPC boards and components support real-time digital status reporting and computer control, and are equipped with instant trip, and true I^2T wire protection. The SSPC and RPC product line has been field proven for military markets, and are used in the Bradley fighting vehicles and M1A2 tank.

DDC is the premier manufacturer of hybrid motor drives and controllers for brush, 3-phase brushless, and induction motors operating from 28 Vdc to 270 Vdc requiring up to 18 kilowatts of power. Applications range from aircraft actuators for primary and secondary flight controls, jet or rocket engine thrust vector control, missile flight controls, to pumps, fans, solar arrays and momentum wheel control for space and satellite systems.

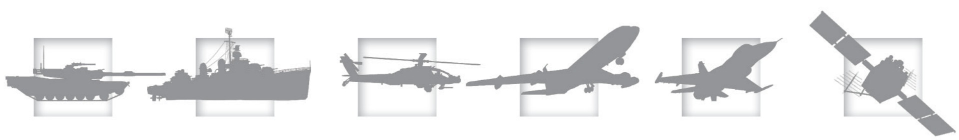
Certifications

Data Device Corporation is ISO 9001: 2008 and AS 9100, Rev. C certified.

DDC has also been granted certification by the Defense Supply Center Columbus (DSCC) for manufacturing Class D, G, H, and K hybrid products in accordance with MIL-PRF-38534, as well as ESA and NASA approved.

Industry documents used to support DDC's certifications and Quality system are: AS9001 OEM Certification, MIL-STD-883, ANSI/NCSL Z540-1, IPC-A-610, MIL-STD-202, JESD-22, and J-STD-020.



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The first choice for more than 45 years—DDC
DDC is the world leader in the design and manufacture of high reliability data interface products, motion control, and solid-state power controllers for aerospace, defense, and industrial automation.

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